T-101 DI water reservoir

A

B

HH trip 9,200 psi

PT 101

P-101 Dual-cylinder syringe pump

0.01-20 mL/min, pulse-free, 10 000 psi, const. flow / const. P

PSV-101 set 9,500 psi

to vent / drain

OV-101 Forced-convection oven, 150 C operating / 177 C design

(alternative: heating jacket on CH-101 only, transfer vessels at room T - compared in Phase 5)

HV-21

HV-22

HV-23

HV-24

TT 101

oven air

TV-101 Brine hydraulic water

TV-102 Crude oil hydraulic water

TV-103 Polymer / surfactant hydraulic water

TV-104 CO2 hydraulic water

HV-11

HV-12

HV-13

HV-14

1000 mL floating-piston transfer vessels, 10 000 psi at 177 C

(TV-104: CO2-rated piston seals, filled by a booster pump, not shown)

manifold with isolation / equalising valves (not shown)

HV-31

PT 102 inlet

PDT-101 A-D 5 / 50 / 500 / 1500 psid

TT 102 holder skin

PT 103 outlet

core 38.1 mm x up to 305 mm

CH-101 Hassler-type core holder, MAWP 10 000 psi @ 177 C

FKM / FEPM sleeve, floating inlet piece with distributor plate, optional dP taps at L/3, 2L/3

HV-32 core bypass

PT 104

PSV-102 set 10,000 psi

RD-101 burst 10,500 psi

PDIC 105

interlock: Pconf - Pin >= 500 psi, else stop P-101

P-102 confining pump

syringe, const. P, 10 000 psi; tracks pore P + net set point

HV-41

confining fluid: water/glycol or silicone oil

dome: N2 regulator or syringe pump

P-103

BPR 101

dome-loaded, 0-10 000 psi, C-276 wetted, heat-traced (CO2 Joule-Thomson cooling)

S-101 sep.

FC-101 fraction collector on balance WT-101

gas meter / GC

GM 101

vent

DAQ / control (PC + logger)

Logs: PT-101..104, PDT-101 A-D, TT-101..102,
pump volumes & rates, WT-101, GM-101

Controls: P-101 rate / pressure; P-102 net confining
set point (tracks PT-102); OV-101 temperature

Trips: PT-101 HH 9,200 psi -> stop P-101
PDIC-105 net confining < 500 psi -> stop P-101
TT-101 HH 170 C -> oven heaters off

Sample rate 1 Hz (10 Hz around breakthrough)

LEGEND

- process line, 1/8 in SS316
- confining line, 1/4 in SS316
- instrument signal
- isolation valve
- check valve
- relief valve / rupture disc

Tube wall per line: see tubing table.
Set points: system/setpoints.json, DECISIONS.md D1.3

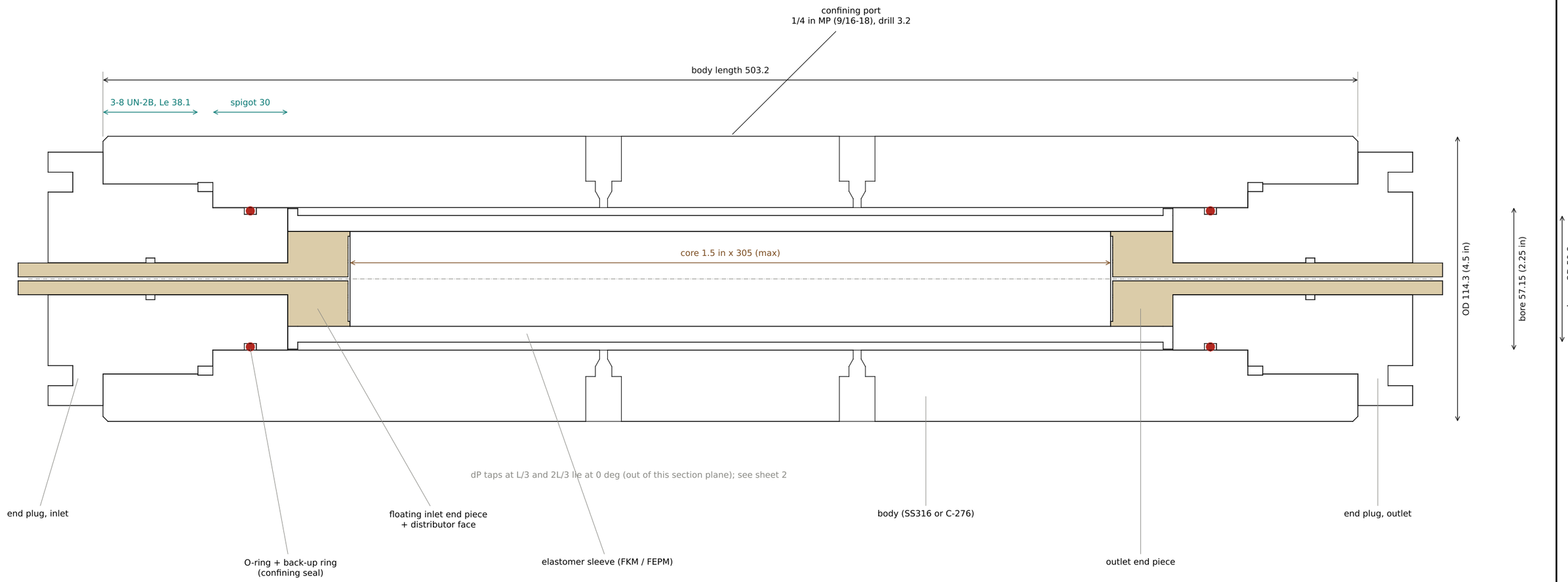
CORE-FLOODING APPARATUS FOR EOR

Piping & instrumentation diagram (simplified)

MPrep Engineering - Case study 04 - Dwg CF-PID-001 Rev A

Typical industry values, not a client specification.

Not for construction. Symbols simplified after ISA-5.1. Generated by system/pid.py



DESIGN DATA

Design pressure (confining)	10 000 psi (68.9 MPa)
Max pore pressure	9 000 psi
Design temperature	177 C (350 F)
Hydrotest (VIII-2 8.2.1)	16 640 psi (SS316) / 14 680 psi (C-276)
Core	38.1 mm (1.5 in) x 25-305 mm; 1 in via adapters
Body material	SA-479 316 (baseline) / SB-574 C-276
End plugs	same alloy as body; anti-galling coating
Sleeve	FKM 75 Shore A, 6.35 wall (FEPM/FFKM for CO2)
Seals	O-ring + PEEK back-up ring
Mass (body + plugs, SS316)	36.4 kg

NOTES

1. Threads shown simplified at nominal diameter. 3-8 UN-2A/2B, 12 threads engaged.
2. Inlet end piece floats axially (stem O-ring) so the core is loaded only by net confining stress.
3. Distributor face: 2 concentric + 8 radial grooves 1.0 x 0.8 mm (baseline; compared in Phase 6).
4. Radial ports: 3.2 mm drill, 59 deg seat cone, 6.6 mm tube bore x 4, 14.3 mm gland counterbore x 18 deep.
5. Break all edges 0.5 x 45 deg; outer ends 2 x 45 deg; bore Ra 0.8 um.
6. Hydrotest per ASME VIII-2 Part 8; marking: MAWP / MDMT / serial.

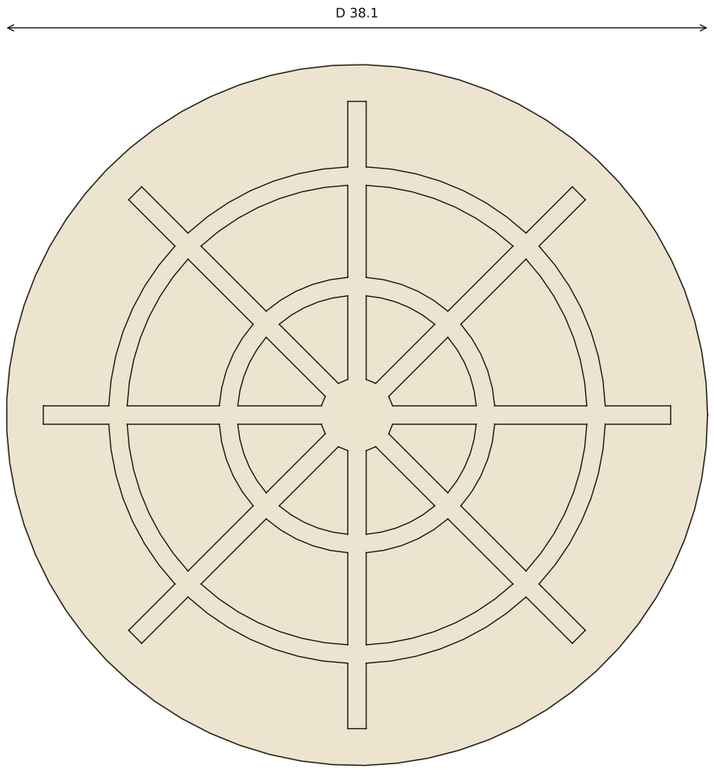
HASSLER-TYPE CORE HOLDER 1.5 in x 12 in

Sheet 1 - assembly, axial section

MPrep Engineering - case study 04 - Dwg CF-CH-001 Rev A - mm - scale as shown

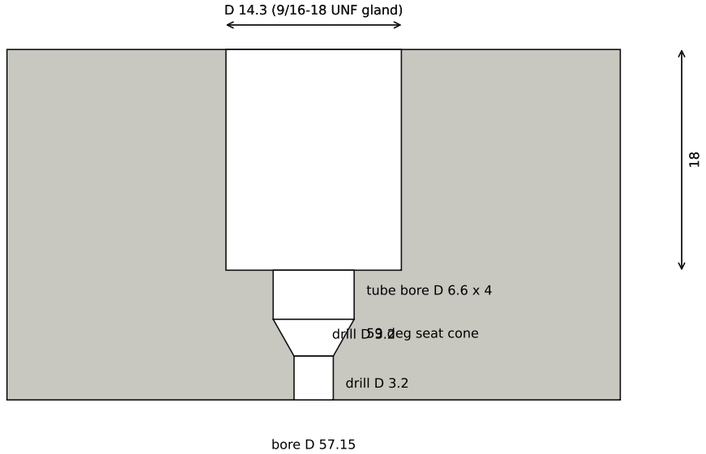
Typical industry values, not a client specification. Not for manufacture.

Generated from the FreeCAD model (cad/build_cad.py -> drawings.py)



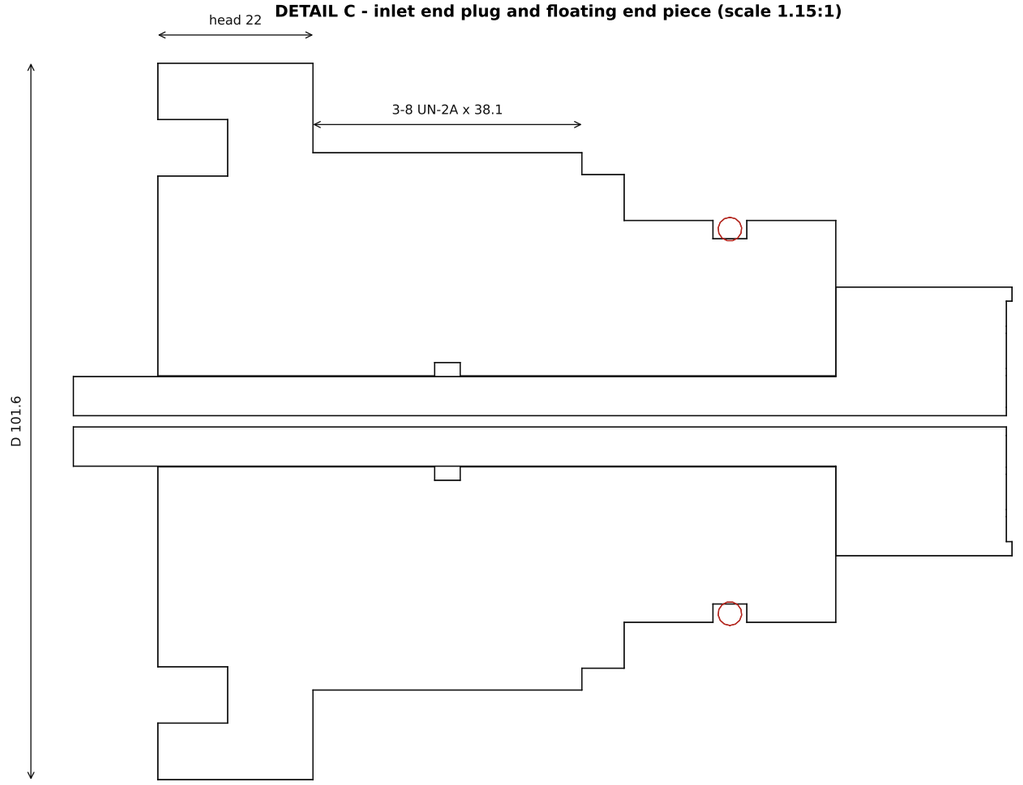
VIEW A - inlet distributor face (scale 3:1)

grooves 1.0 wide x 0.8 deep; rings at r = 7 and 13 mm; 8 radial; centre pocket D 4



DETAIL B - confining port, radial section (scale 2:1). Same detail at the two dP taps (0 deg).

DETAIL C - inlet end plug and floating end piece (scale 1.15:1)



THREAD DATA (ASME B1.1)

3-8 UN-2A (plug) / 3-8 UN-2B (body), major D 76.2 mm, pitch 3.175 mm

Engagement 38.1 mm (12 threads); required >= 15.5 mm (SS316, 0.6 S at 177 C)

Plug load at MAWP: 176.9 kN (P on 57.15 seal diameter)

Anti-galling: dissimilar hardness or Ni-PTFE / silver plating on plug; nickel anti-seize

Dual-start not required; vent groove across threads (leak tell-tale)

HASSLER-TYPE CORE HOLDER 1.5 in x 12 in

Sheet 2 - distributor face, port detail, end plug

MPrep Engineering - case study 04 - Dwg CF-CH-002 Rev A - mm - scale as shown

Typical industry values, not a client specification. Not for manufacture.

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